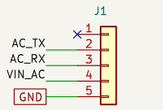
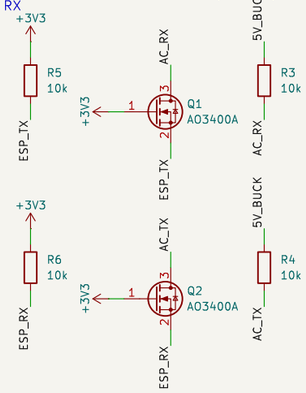
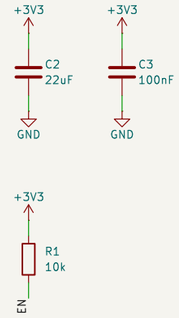
Nerdhome Breeze voor Daikin – S21-port adapter, ESP32-C6 + ESPHome
Plugs into the S21 connector of Daikin indoor units (5 V open-drain UART 2400 8E2).

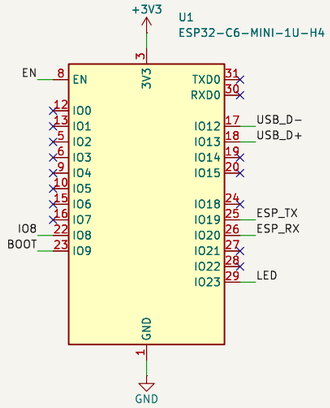
Daikin S21 (JST EH 2.5 mm, EHR-5 cable 1:1) 5 V <-> 3.3 V level shifters (MOSFET), 5 V pull-ups from the buck
1 = Vref 5 V (not used) 2 = AC TX 3 = AC RX
4 = Vcc 12–15 V (board supply) 5 = GND



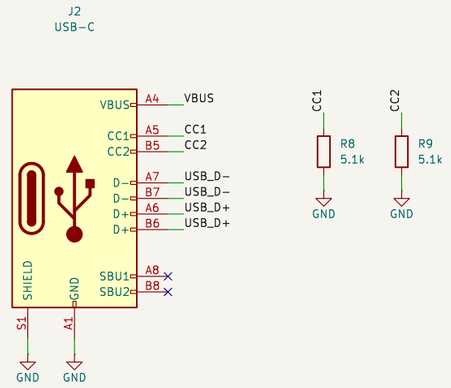




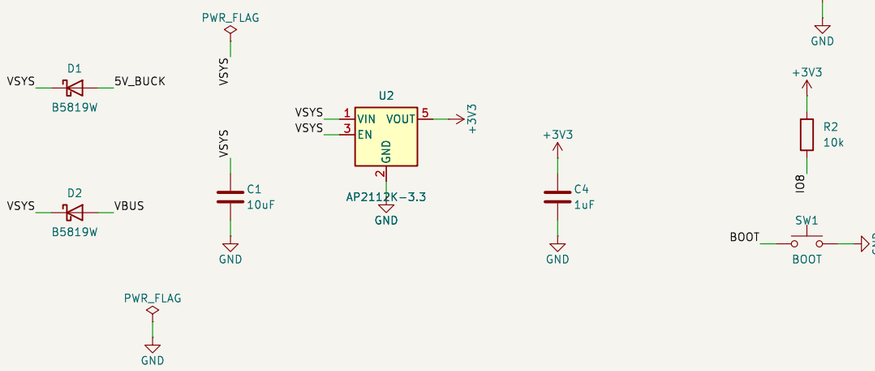
ESP32-C6-MINI-1U (U.FL antenna)
GPIO19 = TX to AC, GPIO20 = RX from AC
GPIO23 = status LED, GPIO9 = BOOT, GPIO12/13 = USB

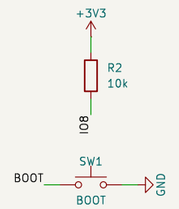


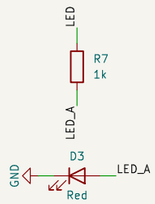
USB-C: flashing + bench power only.
Do NOT plug in USB while the board is connected
to the air conditioner (AC logic ground may not be isolated).



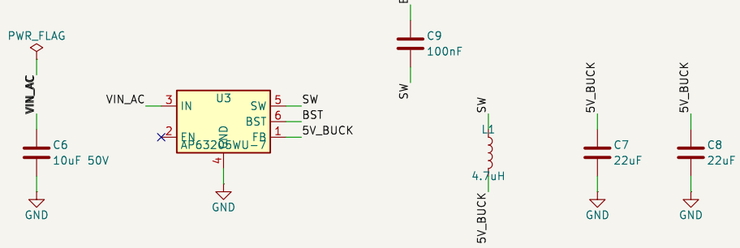
Power: S21 12–15 V -> 5 V buck, or USB 5 V, via Schottky OR (no back-feed), 3.3 V / 600 mA LDO







Buck 12–15 V (S21 pin 4) -> 5 V: AP63205, EN open = auto start
L 4.7 uH, Cin 10 uF/50 V, Cout 2x22 uF, Cbst 100 nF (datasheet table 3)



JLCPCB assembly: LCSC numbers in symbol fields
UART 2400 8E2 @ 5 V open drain on S21; ESPHome daikin_s21 (asund)

Jasper Bom

Sheet: /
File: daikin-c6.kicad_sch

Title: Nerdhome Breeze voor Daikin – S21 adapter (ESP32-C6)

Size: A3	Date: 2026-10-19	**Rev: 1.0**
KiCad E.D.A. 9.0.9		Id: 1/1